



Italian data rescue activity

9 September 2026



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PROGRAMME OF
THE EUROPEAN UNION



IMPLEMENTED BY

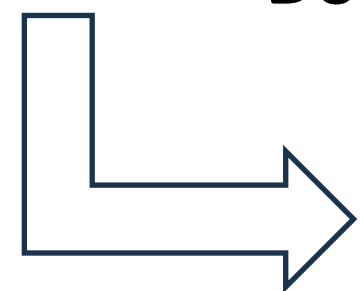


Data rescue activities by ItaliaMeteo - AISAM

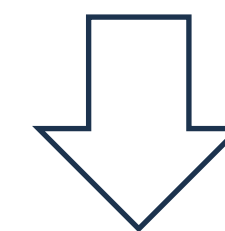
Overview

- The starting point is the dataset created during the «Dieci & Lode» Project (2023-2025), an Italian acronym for “**Climate data of the Former Italian Colonies and their Digitalization**”.
- It includes 40,000 **scanned images** of observations from former Italian territories that were for a period ruled by Italy (particularly, **Eritrea, Somalia, Ethiopia, Libya, and Dodecanese islands, Albania, Dalmatia, Istria**).

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The aims of this NEW Copernicus project are to (i) **organise** and (ii) **start the digitisation** (both with manual keying and the application of OCR-AI/ML techniques) of these historical data records.



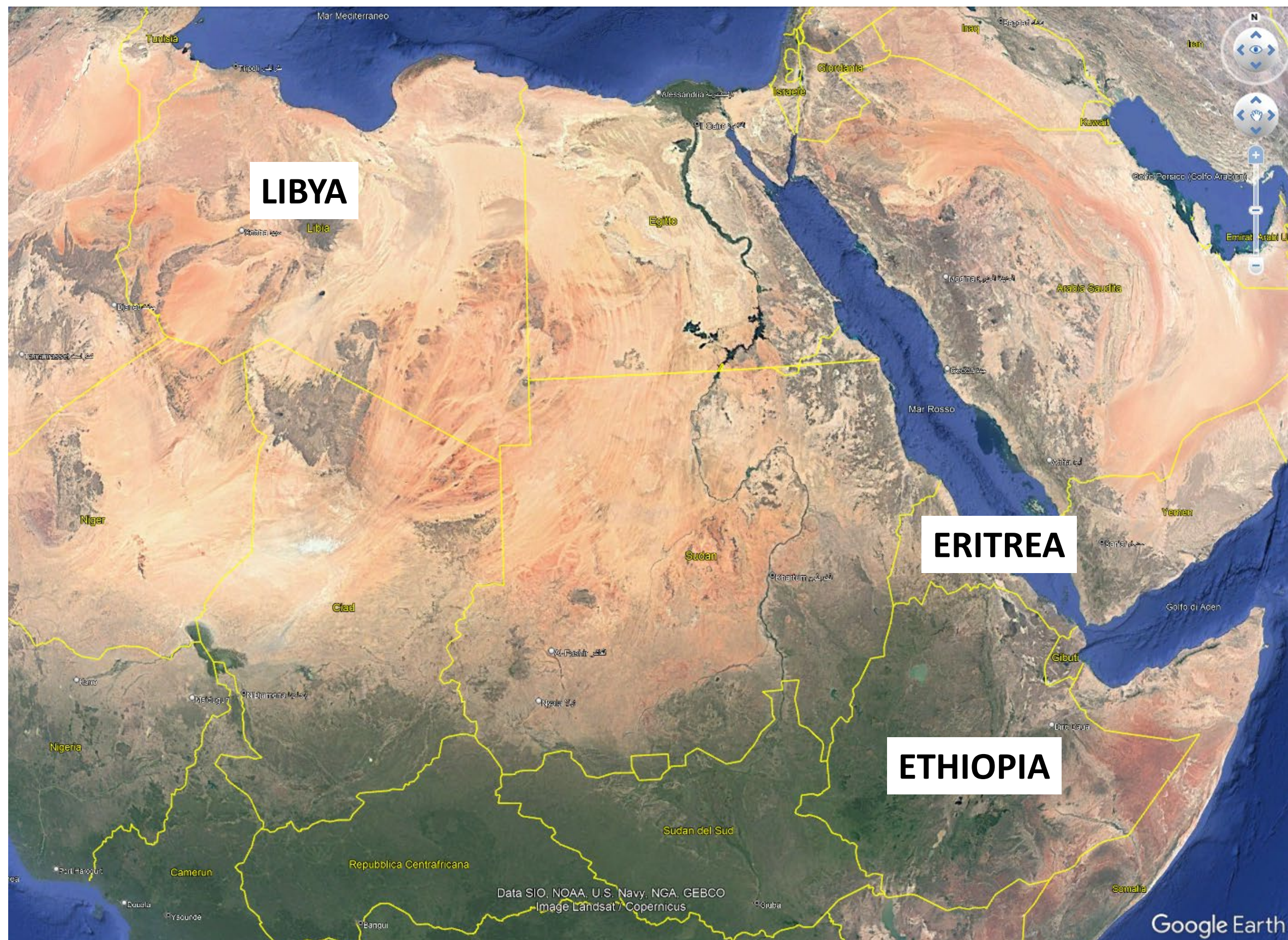
We have started the activities working on two parallel lines:

1. Make a detailed **inventory of documents scanned** by the «Dieci e Lode» project;
2. Build a procedure to **organise the documents and digitise them with OCR**.

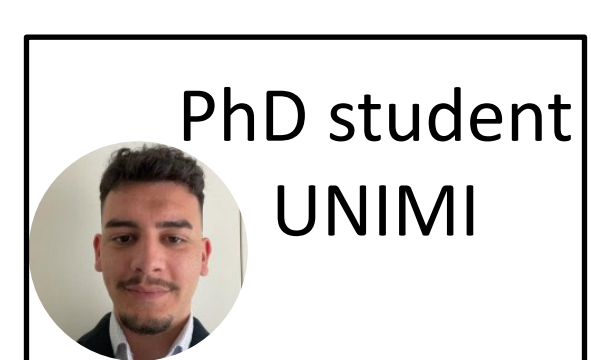




1. Inventory



We have started to make the **inventory**
of
Lybia, Eritrea and Ethiopia.



PhD student
UNIMI



POLIMI master
students



For each country we have prepared an Excel file organised as follow:

STATION	LOCATION	LATITUDE N			LONGITUDE E			ALTITUDE (M A.S.L.)	SOURCE OF DATA COORDINATES	DATA
DEBAROA	Mai Feccia	15	5	15	38	49	55	1930	Book	Total monthly and annual rain
DEBAROA	Mai Feccia	15	5	15	38	49	55	1930	Book	Total monthly and annual rainy days
DEBAROA	Mai Feccia	15	5	15	38	49	55	1930	Book	Relative humidity
DEBAROA	Mai Feccia	15	5	15	38	49	55	1930	Book	Daily precipitation

List of stations with the corresponding coordinates

DATA SOURCES	
AM	Military Aeronautic
B	Book
CR	Crea
WB	Weather bulletins
GSM	
RCO	from NOAA

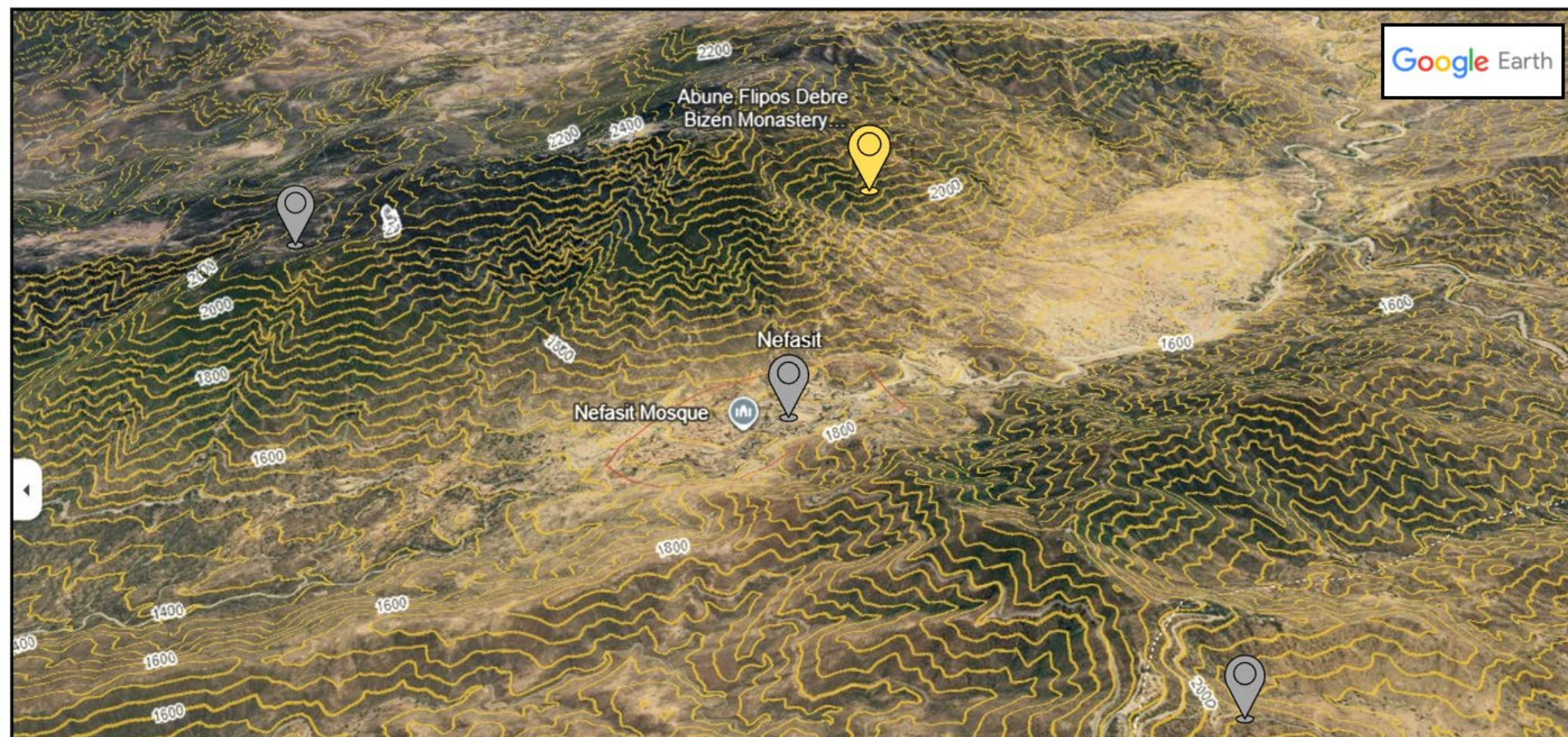
YEAR				
...	1900	1901	1902	1903 ...
	YES (B)	YES (B)	YES (B)	
YES (B)	YES (B)	YES (B)	YES (B)	
YES (AM/CR)	YES (AM)	YES (AM/CR)		

General overview of the data availability for each station and variable



A detailed **quality control of the coordinates** of each station for each country has been done:

- Stations with coordinates: the consistency between latitude, longitude, elevation and the name of the stations has been done using Google Earth → Corrections are made only if a new position is certain.
- Stations with only the elevation: search the station city name with Google Earth and try to identify an approximate location consistent with the reported elevation → It is not easy to pinpoint the exact location, even though the town still exists.



For instance, Nefasit (Eritrea):

The town has lower elevation (1689 m asl) with respect to the reported ones (2082 m asl). The station was probably located in the surrounding area. Therefore, we have selected the point closest to a monastery.

- Stations without the coordinates: search the station city name with Google Earth.

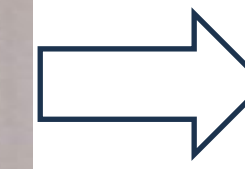


In case of missing coordinates or doubtful coordinates, we search on the web for any additional useful information

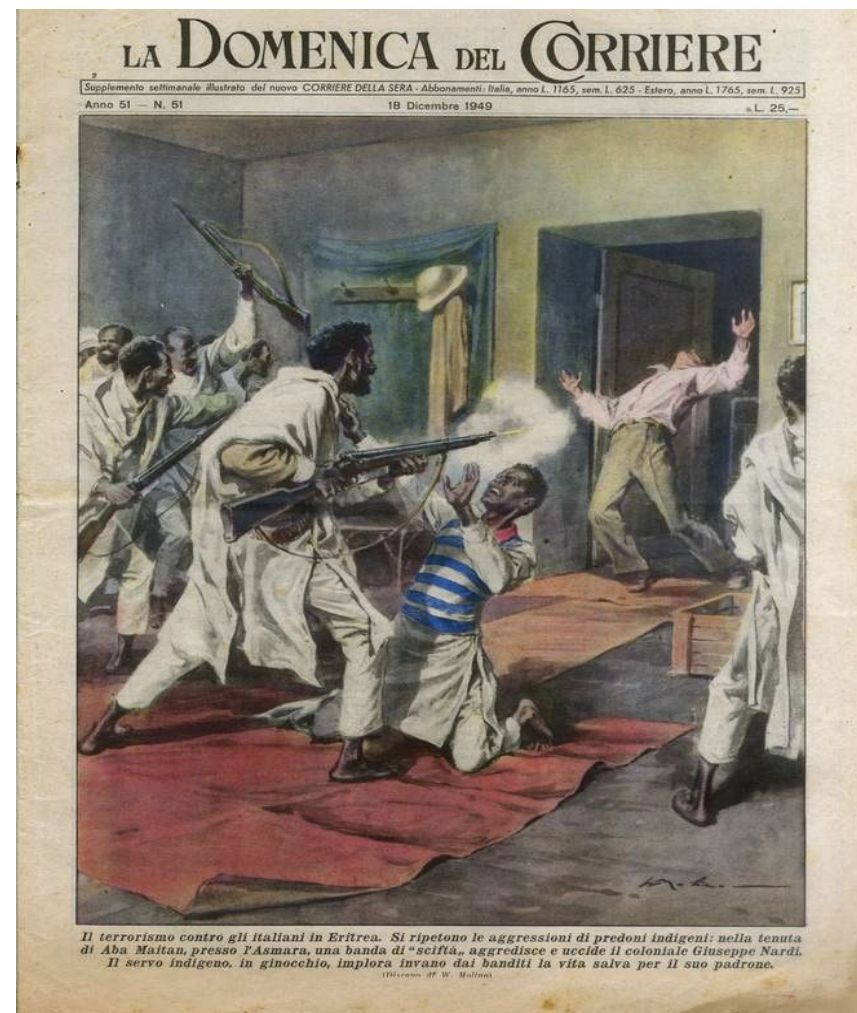
Eritrea

SETTORE CENTRALE
ABBA' MAITAN (Pozzo Ghelebes)
Totali mensili ed annuali delle piogge

Anni	G	F	M	A	M	G	L	A	S	O	N	D	Totale annuo
1930	126.5	61.5	54.5	31.0	0.0	45.5	167.0	77.0	32.0	81.5	0.0	54.0	730.5
1931	85.5	135.0	82.5	42.5	125.5	0.0	67.5	212.0	1.5	89.0	230.0	147.5	1218.5
1932	70.0	117.0	60.5	53.0	89.5	0.0	99.5	233.5	38.0	18.0	18.5	186.0	983.5
1933	150.5	98.5	94.5	98.5	50.0	92.5	72.5	107.0	41.5	97.0	123.5	13.0	1079.0

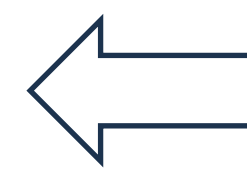


Data from the Aba Maitan weather station, whose coordinates are unknown (source: Book)

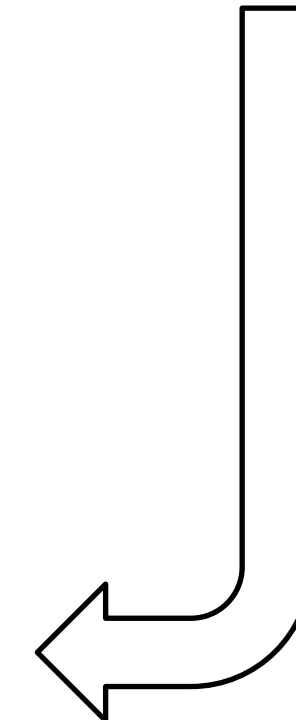


“Violence against Italians in Eritrea. Attacks by local bandits continued: at the **Aba Maitan** estate, near **Asmara**, a group of ‘shifta’ assaulted and killed the settler Giuseppe Nardi. A local servant, kneeling, vainly pleaded with the bandits to spare his master’s life”

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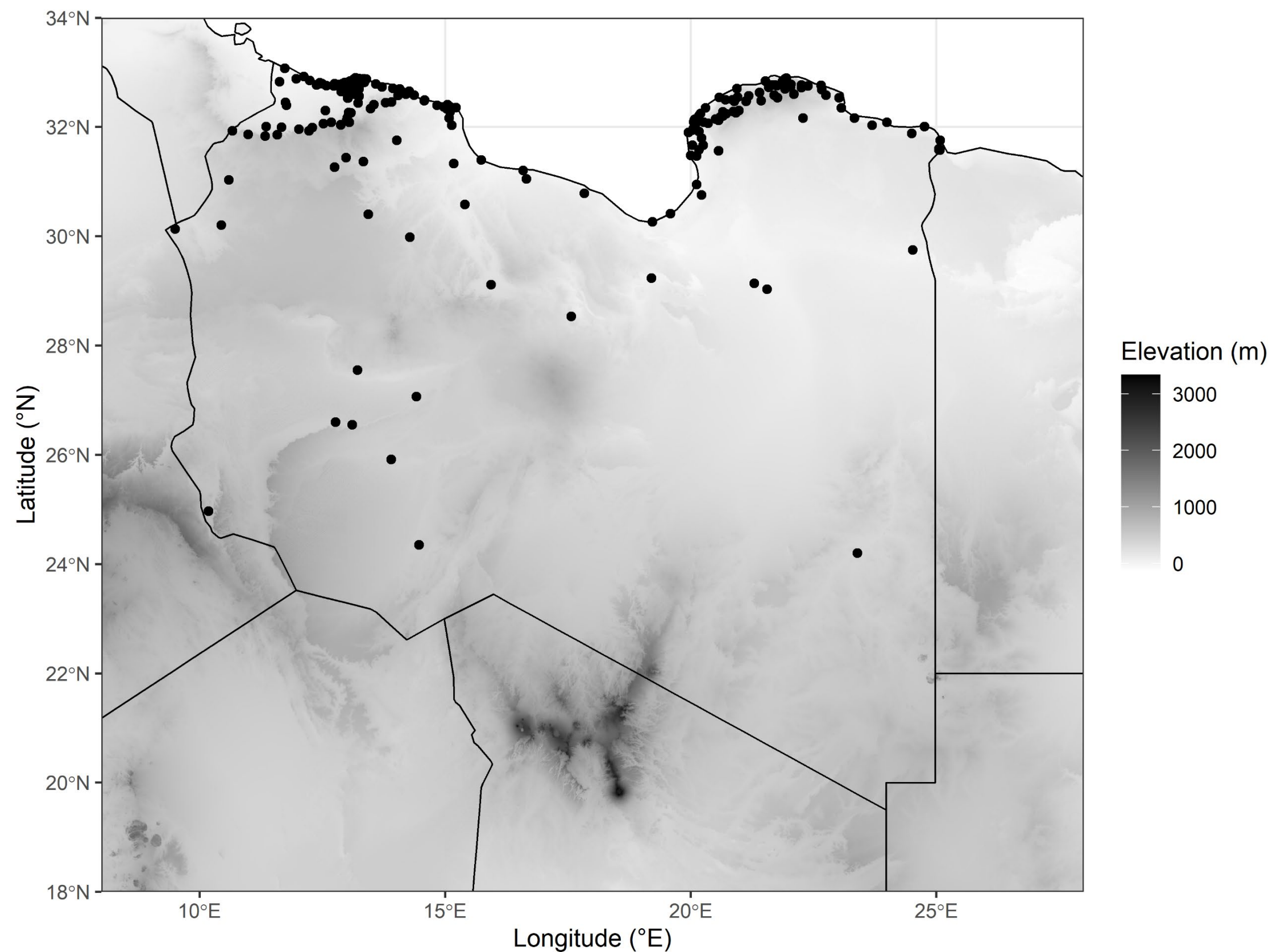
Historical sources



Source: *La Domenica del Corriere*, illustrated supplement of *Corriere della Sera*, Year 51, No. 51, 18 December 1949

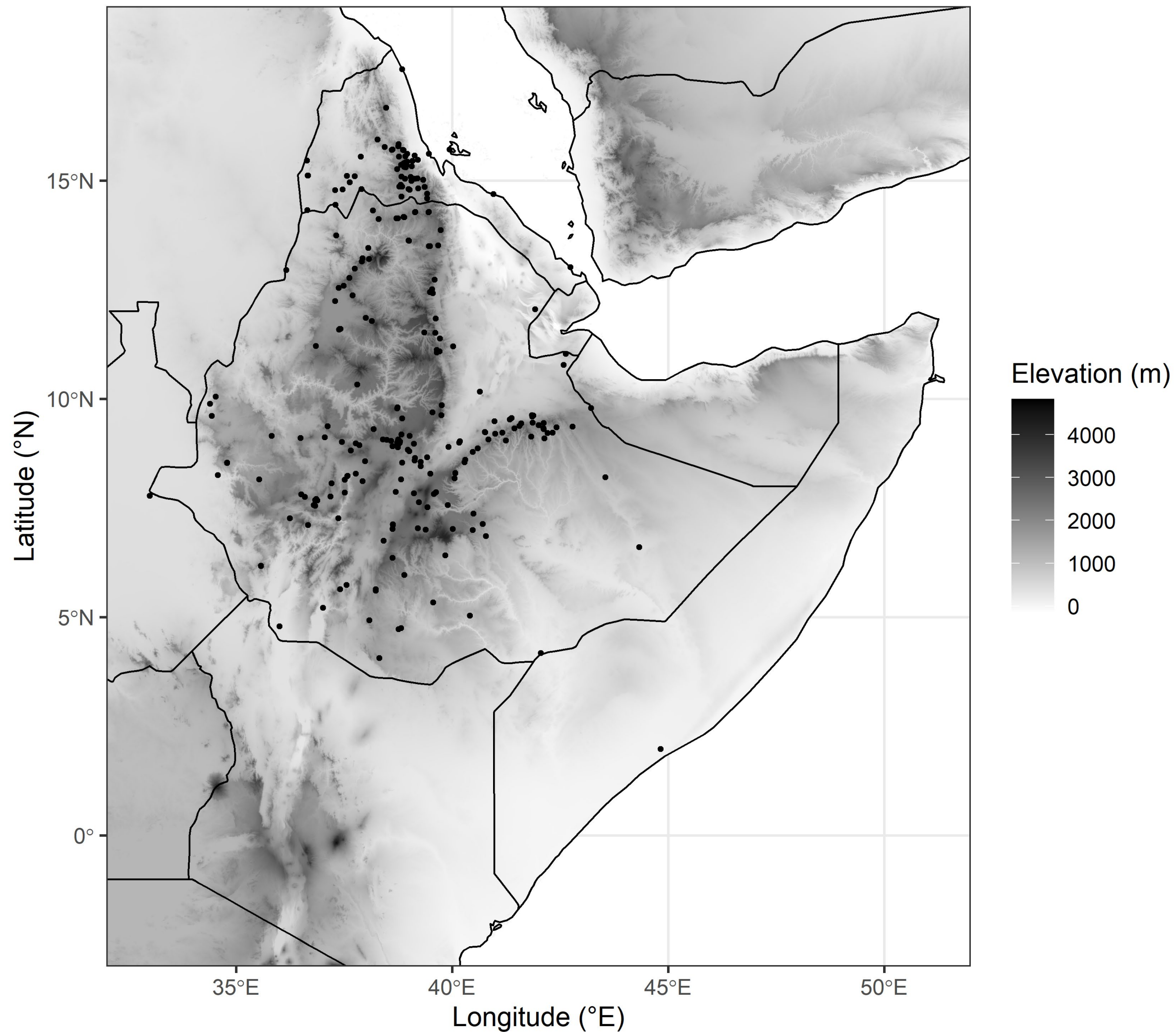


- Today, inventories of Lybia, Eritrea and Ethiopia are completed: all coordinates were checked to verify the consistency between latitude, longitude and elevations;
- The inventory of Somalia is in progress.



Lybia

- 236 stations (10 stations are without coordinates);
- data for the 1879-1944 period;
- different temporal resolution and different variables.



Ethiopia

208 stations (3 stations are without coordinates);

- data for the 1886-1940 period;
- different temporal resolution and different variables.

Eritrea

105 stations (11 stations are without coordinates);

- data for the 1881-1960 period;
- different temporal resolution and different variables.



2. Comparison with other datasets

- We are downloading and checking the data already available online for Lybia, Eritrea, Ethiopia and Somalia:



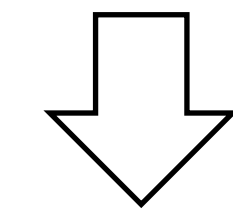
We have started with the subdaily, daily and monthly data from the Copernicus

- In addition, we are checking information contained in our archives for Ethiopia and Eritrea with the inventory provided by the Maynooth University
- Ethiopia → there is no overlap for any of the stations identified, because our time series end in 1940, whereas the Simon stations begin in the 1950s
- Eritrea → there is overlap for some stations. It will be necessary to check station by station, and variable by variable.

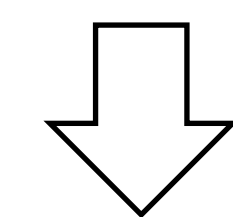


3. Cropping and OCR software

We have decided to start the **automatic digitisation** with the **Books**, because **they include printed data**. The organisation of the tables is very similar even if they contain different variables.

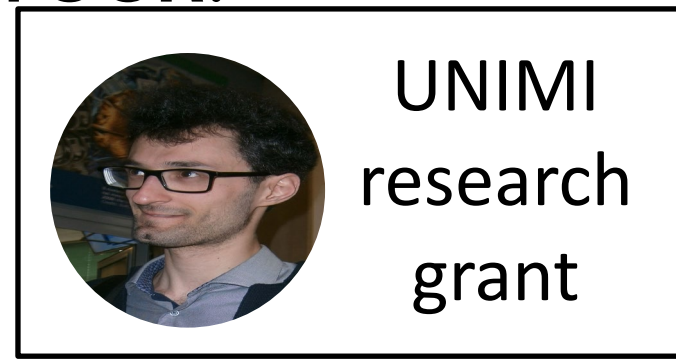


6 volumes concerning Cirenaica (East Libya), Tripolitania (North Libya), Libya Somalia, Eritrea, Ethiopia



We are dividing the activity in **two parts**:

- Set up a **procedure to cut and prepare the tables** → This is useful for OCR, but it also better organises the material and makes it accessible for consultation (this will be partially manual and partially automatic).
- Prepare a procedure to **digitise the data with OCR**.



ZONA CLIMATICA Clima marittimo TRIPOLI Osservatorio Centrale della Libia (Lat. 32°54'00"N; Long. 13°10'33"E.G.; altitud. s.l.m. 22 m) Località: Castello (Bastione NW) Pressione barometrica ridotta a 0° temperatura (prima cifra mb 10)													
Anni	G	F	M	A	M	G	L	A	S	O	N	D	Media annua
1893	?	?	14.4	13.5	11.1	12.3	11.9	14.2	14.2	14.8	12.4	17.3	(13.6)
1894	14.7	13.5	12.4	10.6	10.0	14.4	13.9	14.0	13.9	13.8	15.0	13.4	13.3
1895	12.0	09.9	11.0	11.6	13.0	13.2	13.6	13.1	16.7	13.2	18.2	13.8	13.3
1896	15.9	17.9	12.0	12.7	09.4	12.0	13.4	11.9	12.8	12.4	12.2	13.1	13.0
1897	11.9	19.9	15.8	12.0	09.9	13.5	11.4	12.2	14.3	15.0	19.6	17.1	14.3
1898	21.6	12.8	07.6	11.2	11.4	12.1	13.1	14.3	14.3	11.8	13.1	18.0	13.4
1899	18.7	15.2	13.1	13.8	12.4	11.8	14.3	13.5	13.1	16.2	18.3	13.2	14.5
1900	13.5	13.4	11.6	13.5	10.3	12.7	13.2	11.6	15.9	16.4	12.4	18.7	13.6
1901	15.9	10.2	14.3	13.5	10.7	13.8	11.4	12.6	10.2	11.9	15.4	15.1	13.0
1902	21.6	12.2	12.6	11.4	13.6	10.6	13.7	13.9	13.2	11.6	13.2	16.2	13.6
1903	20.7	24.0	17.8	09.5	09.2	12.0	13.2	13.4	13.9	14.0	15.2	(13.2)	14.7
1904	16.2	13.1	10.3	12.4	16.3	13.2	13.8	15.0	12.8	13.9	17.2	17.8	14.3
1905	19.5	18.7	15.1	12.2	?	?	13.4	14.0	14.2	?	?	?	?
1906	(18.4)	(12.6)	(16.2)	(13.9)	(11.8)	(12.6)	(14.0)	(15.2)	(15.8)	(14.6)	(17.0)	(15.0)	(14.8)
1907	(19.9)	(11.2)	(17.5)	09.1	12.7	14.3	13.0	14.4	13.9	13.6	16.4	19.5	14.6
1908	19.2	19.0	12.6	11.0	15.4	14.3	13.9	14.3	16.7	(16.8)	(17.0)	15.2	15.4
1909	17.4	13.1	09.4	13.2	12.0	13.6	13.8	13.4	13.5	14.8	14.0	16.2	13.7
1910	16.4	15.4	15.9	12.2	09.4	13.0	(13.0)	14.2	14.4	17.1	17.2	14.0	14.3
1911	16.6	19.8	11.8	12.3	10.6	15.6	?	?	?	?	?	?	?
1912	17.5	18.0	18.3	11.8	15.2	12.6	12.8	12.7	14.2	15.6	15.5	21.4	15.5
1913	18.7	15.0	18.0	11.5	12.3	15.6	13.2	13.4	(13.9)	14.7	19.8	17.9	15.3
1914	14.7	15.6	15.1	16.2	13.9	11.9	11.6	13.4	16.3	13.4	09.8	18.3	14.2
1915	12.7	16.8	13.6	11.9	12.4	11.4	12.0	12.0	13.0	?	?	?	?
1916	17.2	11.4	09.6	11.0	12.3	10.4	12.8	12.2	?	?	12.0	?	?
1917	10.0	12.6	10.2	13.1	13.8	12.2	?	11.4	15.5	13.1	12.4	14.2	(12.6)
1918	21.4	20.0	12.6	10.0	13.2	?	12.3	15.6	14.5	12.2	12.7	15.4	(14.5)
1919	08.9	?	15.9	14.4	15.8	17.1	15.0	19.4	16.0	16.7	17.0	18.6	(15.9)
1920	21.5	21.8	17.1	15.6	15.6	14.0	15.6	14.4	17.0	13.0	16.2	18.0	16.6
1921	21.8	17.8	21.4	11.0	13.5	15.6	(13.2)	?	16.4	18.8	16.0	17.4	(16.6)
1922	16.0	17.1	18.6	15.1	16.0	13.1	15.1	14.0	13.6	13.1	15.0	17.9	15.4
1923	15.6	14.7	14.2	10.4	13.4	13.9	15.4	13.4	16.2	17.2	14.8	17.5	14.7
1924	15.2	11.9	12.2	13.8	15.0	13.6	13.0	12.8	14.7	15.6	16.6	17.6	14.3
1925	27.5	19.0	11.8	13.2	10.6	11.8	13.5	14.2	14.0	13.0	11.8	19.6	15.0
1926	17.5	17.2	14.3	13.5	11.4	11.9	13.4	15.0	16.4	16.3	14.7	16.3	14.8
1927	17.4	18.2	15.5	14.3	12.6	13.8	12.2	14.0	13.8	15.8	16.2	12.3	14.7
1928	16.4	20.2	12.2	10.2	10.7	13.8	15.5	13.1	14.0	15.2	15.5	17.0	14.5
1929	19.6	12.2	16.4	13.0	12.2	14.4	15.9	13.0	15.0	16.7	15.4	19.1	15.2
1930	17.5	14.6	14.7	09.4	14.8	12.4	12.2	15.9	15.8	16.6	19.8	14.8	14.9
1931	15.0	12.4	13.4	11.8	13.8	15.2	13.6	11.8	13.9	16.4	15.5	17.2	14.2
1932	23.5	15.2	13.0	14.3	14.4	13.0	11.0	15.4	13.8	13.0	16.8	20.2	15.3
1933	14.7	11.8	14.2	14.8	12.7	10.2	15.2	14.4	14.8	15.4	12.0	12.4	13.5
1934	19.6	17.8	10.2	10.8	12.7	13.0	13.0	11.5	15.5	14.8	14.0	15.6	14.0
1935	14.3	16.2	14.3	13.5	10.8	15.1	13.8	12.4	14.2	13.5	14.8	12.7	13.8
1936	16.6	09.6	12.0	11.2	08.6	13.6	13.5	14.6	15.9	13.4	12.8	18.2	13.3
1937	19.0	12.8	11.9	10.7	13.2	11.8	11.9	10.7	11.8	12.4	13.8	13.6	12.8
1938	15.6	17.0	19.5	11.0	11.5	14.4	12.7	09.8	12.4	13.4	15.9	10.4	13.6
1939	13.2	12.0	11.6	09.6	09.2	11.4	13.9	12.8	12.3	14.0	16.5	16.0	12.7
Media	17.1	15.3	13.9	12.3	12.4	13.2	13.4	13.5	14.4	14.5	15.2	16.2	14.3
Valori estremi delle medie mensili di tutto il periodo													
Max.	27.5	24.0	21.4	16.2	16.2	15.6	15.9	15.9	17.0	18.8	19.8	21.4	1027.5
Min.	11.9	09.6	07.6	09.1	08.6	10.2	11.0	09.8	10.2	11.6	09.8	10.4	1007.6



- One
sheet for
30 days

MINISTERO DELL'AGRICOLTURA E DELLE FORESTE

MODELLO D. 39

R. UFFICIO CENTRALE DI METEOROLOGIA E CLIMATOLOGIA - ROMA

SERVIZIO TERMO-PLUVIOMETRICO

13 FEB

Stazione di **R. STAZIONE SEGNAI - BENGASI**
Provincia di **Benevento**
Bacino di **...**

Altazza della stazione sul mare, m. **20**
Altazza dei termometri sul suolo, m. **175**
Altazza del pluviometro sul suolo, m. **0.75**

Anno 1940 - **XVII** E. F.

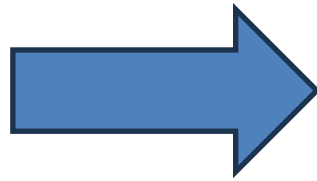
Mese di **febbraio**

Le osservazioni vengono effettuate alle ore **19.00**

Giorni	Stato del cielo	Direzione e forza del vento a stima	TEMPERATURA			PRECIPITAZIONI						STATO del mare
			Massima	Minima	Media diurna	Durata del fenomeno	Altezza in mm.	Forma	Altezza del vento in mm.	Altezza del vento in mm.	Altezza del vento in mm.	
1	2	3	4	5	6	7	8	9	10	11	12	
1	M.	S.W. 1	17.5	15.0	16.2	Calda e discesa la notte	2,2	p.				X
2	C.	S.W. 2	17.6	14.2	16.2							X
3	M.	N. 1	17.5	15.2	16.3							X
4	S.	S. 1	17.2	14.2	14.2							X
5	M.	S. 2	18.0	13.0	16.5							X
6	C.	W. 2	18.2	13.8	16.0							X
7	C.	N.W. 4	18.4	12.1	15.2	Calda nella notte	0,9	p.				X
8	M.	N. 2	15.8	12.8	14.3	Calda nella notte	3,2	p.				X
9	C.	N.W. 0	15.9	13.9	14.9							X
10	S.	S. 2	15.6	11.2	13.4							X
Somma I Decade			175.7	133.0	153.2		38.3					
Media I Decade			17.5	13.3	15.3							
11	M.	S. 1	17.0	12.0	14.5							X
12	M.	N. 1	16.2	13.9	15.0							X
13	M.	S.E. 1	17.5	10.0	13.7							X
14	M.	N.E. 1	17.4	8.8	13.1							X
15	M.	S.E. 3	16.0	10.0	13.0							X
16	M.	W. 3	16.5	10.0	13.4	Durata il giorno	1	p.				X
17	M.	S.W. 2	16.5	12.8	14.6	Calda nella notte	1,5	p.				X
18	M.	W. 2	16.8	11.6	14.2							X
19	C.	N.W. 2	17.0	9.0	13.0							X
20	M.	S. 2	16.5	9.1	14.9							X
Somma II Decade			167.4	107.5	139.4		2.5					
Media II Decade			16.7	10.7	13.9							
21	S.	S. 1	17.0	9.0	13.0							X
22	M.	N.E. 0	19.6	10.4	15.0							X
23	S.	S.E. 4	20.4	12.4	16.0							X
24	M.	W. 3	20.4	12.4	16.5							X
25	M.	S.W. 4	19.6	13.4	16.5	Calda nella notte	1,3	p.				X
26	C.	S.W. 3	15.5	10.5	13.0	Altezza 24 ore	5,6	p.				X
27	C.	N. 2	14.9	12.8	13.8	nella 24 ore	3	p.				X
28	C.	N.W. 3	15.2	10.6	12.9	Durata il giorno	0,4	p.				X
29	C.	N.W. 3	16.3	14.0	15.1	Durata il giorno	5,9	p.				X
30	C.	E. 0	16.5	12.8	14.1	Altezza 24 ore		p.				X
31	C.	S. 1	15.4	12.2	13.5							X
Somma III Decade			191.1	131.3	159.4		16.5					
Media III Decade			19.1	13.1	15.9							



One sheet for 10 days



- About 4,700 pages;
- The template is stable over the years and stations;



ZOONIVERSE

One sheet for 30 days



- About 270 pages;
- The template is not stable over the years and stations;



We are evaluating which is the best way: research team/students



Thank you for the attention

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